

Task 1 – Question Answering Task (Report Submission)

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Artificial Intelligence and Machine Learning

(6CS012)

Title: Task 1 Question Answer

(Report)

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Course : BSc (Hons) Computer Science

Tutor : Ms. Jinu Nyachhyon

Submitted on : 10 May 2026

Part I: Question and Answer

Long Question (Chosen: Question 2) [5 Marks]

Ans: 1. Data Drift and Model Degradation

Data drift is when the statistical characteristics of the incoming data change with time, and a model built on the historical data becomes less accurate with each new data sample. This can be seen in an e-commerce environment as changing customer buying patterns around seasonal events or economic fluctuations. When not handled properly, model degradation can result in sub-optimal recommendations, lower conversion rates, and ultimately lost revenue. A helpful approach is to use automated data monitoring pipelines to monitor distributions of features and to retrain if a certain amount of drift is detected. It is possible to identify drift at an early stage using tools like Evidently AI or custom statistical tests (e.g. KL divergence, PSI). The cost and complexity trade-off is the need for a more complex infrastructure and the requirement for more compute resources for ongoing training.

2. Latency and Scalability in Real-Time Inference

For e-commerce platforms, that means millions of users with low-latency predictions, like product rank personalization or dynamic pricing. If latency is not managed, it will lead to poor customer experience and higher bounce rate. Some practical solutions are: model quantization, serving optimised model formats (such as ONNX, TensorRT), and using scalable APIs (such as containers orchestration platforms like Kubernetes) to deploy models behind. Horizontal scaling and load balancing help provide throughput during traffic spikes. The compromise between model accuracy and inference speed must be taken, whereby smaller and faster models could compromise in predictive performance.

3. Monitor and observe the production models

If not monitored, silent model failures can persist for a long time. The scores for predictions, output distributions, business KPIs and other metrics all must be monitored simultaneously. Integrating logging systems (such as MLflow, Prometheus, Grafana) and automated alerts are some of the solutions. Safe validation of new model versions allows safe testing of deployment for new versions, including shadow deployment and A/B testing. The cost comes with the engineering effort needed to build and maintain a healthy observability infrastructure.

4. Cross-Functional Collaboration

No reliable ML system can be constructed alone. Data scientists set the goals for the model and its evaluation, engineers build scalable serving infrastructure, and product

managers make sure that the ML capabilities meet the business requirements. A shared definition of success metrics, regular cross-team alignment and shared dashboards ensure that model improvements have a measurable business impact. One of the most frequent causes of production problems is the lack of communication among these positions.

5. The role of the modern tools and automation

Modern MLOps platforms like Kubeflow, SageMaker, and Vertex AI automate retraining pipelines, model versioning, deployment pipelines, and much more, drastically cutting down manual intervention. Powered by LLM capabilities, emerging systems also provide for automated documentation generation, anomaly explanation, and intelligent alerting, enabling teams to respond quickly to production issues with a higher context understanding.

Short Question - Category 1: Overfitting (Chosen: Question 2) [2.5 Marks]

Ans: 1. Dropout

A percentage of neurons are randomly turned off for each training forward propagation. The idea is that the network will learn more distributed and robust representations by not allowing any individual neuron to become too reliant on any one input. At training time, this adds noise to the system, providing implicit regularisation and at inference time, all neurons are active, but their weight is scaled to match. In image classification, for instance, detecting skin lesions in medical imaging, dropout after dense layers helps to avoid overfitting towards pixel patterns, offering more general visual features.

2. L2 Regularization (Weight Decay)

In L2 regularization, a weight penalty term is included in the loss function which is proportional to the sum of the squared weights. The idea behind the intuition is that if a model has large weights, then it is depending too much on certain features in a dataset, and you want to discourage this so that the model will spread learning more evenly. The resulting decision boundaries are smoother and the generalization is improved. For example, in fraud detection applications, L2 regularization helps to prevent the model from relying too heavily on a single variable in the transaction, such as its amount, and instead focuses on a wider range of information to make decisions about whether it is fraudulent or legitimate.

Short Question - Category 2: Neural Network Architecture (Chosen: Question 1) [2.5 Marks]

Ans: Input Structure and Data Type

CNNs are appropriate for data that has a spatial structure, such as an image organized into a grid of pixels, in which local features like the shape of an edge, or texture, are

meaningful. By contrast, RNNs are used for sequential data, where the order of events is important, e.g., time series sensor readings, text or speech.

How information flows and how information is organised.

A CNN captures information at different levels of the hierarchy and through convolutional and pooling layers in a forward direction. No time or ordering concept between the inputs. An RNN will have a hidden state that can carry information from the previous timesteps of the sequence, in addition to the forward flow of information.

Parameter Sharing

CNNs have filters where weights are shared across space, so they are translation-invariant and can be applied efficiently to image data. All weights in RNNs are shared across time steps, which is like using the same transformation for each position of the sequence. CNNs are suitable for image classification problems like handwritten digit recognition (MNIST) and RNNs for sequential problems like predicting the next word in a sentence or forecasting stock prices.